

Alessandro Porcheddu

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Education

University of Bologna

B.SC. IN COMPUTER SCIENCE AND ENGINEERING

Cesena, Italy

Sept. 2023 - Ongoing

- **Relevant Coursework (4.0 GPA):** Algorithms and Data Structures (**C**), Object-Oriented Programming (**Java**), Linear Algebra for Machine Learning (**Python**), Programming (**C**)

Work Experience

UniBo Motorsport - Motostudent Team

E-POWERTRAIN DIVISION MEMBER

Bologna, Italy

Sept. 2025 - Ongoing

- Currently developing a **custom firmware** for the motorcycle's inverter.
- Cooperating in the implementation of the telemetry through a custom **Grafana** dashboard.

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SYSTEM ADMINISTRATOR INTERN

Cesena, Italy

Sept. 2025 - Ongoing

- Configured and maintained Windows **Domain Controller**, ensuring reliable system performance and uptime.
- Documented system configurations, network layouts, and **maintenance procedures** for internal use.
- Worked with virtualization tools such as **VMware and Proxmox** to create and test system environments.

Cables

AUTOMATION TECHNICIAN

Cesena, Italy

Jun. 2023 - Jun. 2023

- Assisted in assembling electrical control panels for **industrial automation systems**, following detailed wiring diagrams and specifications.
- Troubleshoot **PLC** input/output issues and performed routine checks on control system hardware.

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IT ASSISTANT

Wrocław, Poland

Jun. 2022 - Jul. 2022

- Completed the **Erasmus+** "Trainee for Industry 4.0 Plus" internship program, gaining practical experience in industrial environments.
- Assisted with the digitization of paper records by transferring them to solid-state storage systems.

Side Projects

JavaDyno (Engine Dynamometer Management Software)

 Code

SOFTWARE DEVELOPMENT

- Developed an engine dynamometer management system (EDMS) using **JUnit** automated testing and **JavaFX**.
- Implemented an **MVC** core structure, enabling data exportation and session logging capabilities.
- Ensured **OBD-II** over **CAN** compatibility for in-vehicle testing, enhancing real-world applicability.
- Integrated **WebSocket** and **Serial** connection support for versatile device communication and included simulation features.

Motorcycle Parts E-commerce Platform

Live

WEB DEVELOPMENT

- Built a full-stack e-commerce platform using **Laravel 12**, **React 19**, and **Inertia.js**.
- Implemented secure payment processing with **Stripe** via **Laravel Cashier**.
- Developed an **admin dashboard** for product, order, and customer management with analytics and reporting tools.
- Ensured robust security with mandatory email verification, protected routes, and **Stripe webhook signature validation**.

Motorcycle Exhaust Valve Control Unit

 Code

EMBEDDED SYSTEMS

- Developed an electronic exhaust valve control unit based on **ESP32**.
- Created a browser app to control the device via WiFi through WebSocket communication.
- Designed a custom PCB using **KiCad** and a custom 3D printed enclosure using **Fusion360**.
- Implemented **secure OTA** updates through a custom remote server based on Node.js.

Skills

Technologies Java, C, C++, TypeScript, React, Python, PHP, Cuda, AWS, KiCad, Fusion360, Matlab, , CAN-Bus, IoT, JTAG, RTOS, VMware

Languages Italian, English